UNIT – IV

Eigenvalues, Eigenvectors and Diagonalization of Matrices

Eigenvalues and Eigenvectors:

A matrix eigenvalue problem considers the vector equation

$$AX = \lambda X \qquad (1)$$

Here A is a given square matrix of order n, λ an unknown scalar, and X an unknown vector of order $n\times 1$ (i.e, column vector). In eigenvalue problem, the task is to determine $\lambda's$ and X's that satisfy (1). Since $X = 0$ is always a solution for any λ and thus not interesting, we only admit solutions with $X \neq 0$. The solutions to (1) are given the following names:

- the $\boldsymbol{\lambda}$**'**s that satisfy (1) are called **eigenvalues** of A.
- the corresponding nonzero **X's** that also satisfy (1) are called **eigenvectors** of A.

Thus eigen values are scalars λ which satisfy the equation (1) , i.e, $\boldsymbol{AX} = \boldsymbol{\lambda X}$ for $\boldsymbol{X} \neq \mathbf{0}$**,** we can write

$$(\boldsymbol{A} - \boldsymbol{\lambda I})\mathbf{X} = \mathbf{0} \qquad (2)$$

Equation (2) represents a system of n homogeneous equations in the n variables $x_1, x_2, \ldots, x_n$. Equation (2) has non-trivial solutions (i.e, $X \neq 0$) if the coefficient matrix (A − λI) is singular.

$$\text{i.e., } |\mathbf{A} - \boldsymbol{\lambda}\mathbf{I}| = \mathbf{0} \qquad (3)$$

Expansion of the determinant gives a nth degree polynomial $P_n(\lambda)$ known as the **Characteristic polynomial** of A. Therefore, equation (3) is known as the **Characteristic Equation** of A. Thus, eigen values of square matrix A of order n are the roots of the Characteristic equation (3). Hence, the eigenvalues of A are also called as **Characteristic values of** A. Also, the matrix A can have at most n eigen values.

Degree of a characteristic polynomial = Order of the matrix A.

Spectrum of A is the set of all eigen values of A. The eigen vector X corresponding to an eigen value λ is obtained by solving the homogeneous system (3) with this known eigen value λ.

Eigenvalue problems come up all the time in engineering, physics, geometry, theoretical mathematics, biology, environmental science, urban planning, economics and other areas.

Properties of eigenvalues and eigenvectors:

1. If A is real, its eigen values are real or complex conjugates in pairs.
2. Trace of A = sum of the eigen values of A.
3. The determinant of A = product of eigen values of A.
4. A and A^T has same eigen values.
5. A^m has eigen values λ^m, m being an integer.
6. If λ is an eigen value of A, then $\lambda \pm k$ is an eigen value of $A \pm kI$, k being an integer.
7. If λ is an eigen value of an orthogonal matrix A then $\frac{1}{\lambda}$ is also an eigen value of A.
8. If all the eigen values are non-zero, then $|A| \neq 0$ since |A| = product of the eigen values. i.e., A is non-singular.
9. If at least one eigen value is zero, then |A| = product of eigen values $\lambda_1.\lambda_2...\lambda_n = 0$ i.e., A is singular.

Solved Problems

Example 1: Find the eigen values and eigen vectors of

$A = \begin{pmatrix} 8 & -4 \\ 2 & 2 \end{pmatrix}$ **[Problem related to properties- 1, 2 and 3)]**

Solution: The eigen values are the roots of the **characteristic equation**

$$\begin{vmatrix} 8-\lambda & -4 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (8-\lambda)(2-\lambda)+8=0$$

$$\Rightarrow \lambda^2 - 10\lambda + 24 = 0$$

$$\Rightarrow (\lambda-4)(\lambda-6)=0$$

The two distinct **eigen values** are λ = 4, 6.

Let $X_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be the eigen vector corresponding to eigen value λ = 4:

Then $(A-\lambda I)X_1 = 0$

$$\Rightarrow \begin{pmatrix} 8-4 & -4 \\ 2 & 2-4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 4 & -4 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \Rightarrow \begin{matrix} 4x_1 - 4x_2 = 0 \\ 2x_1 - 2x_2 = 0 \end{matrix}$$

$$\Rightarrow x_1 = x_2 \quad \therefore X_1 = C_1\begin{pmatrix}1\\1\end{pmatrix},\ C_1 \text{ is arbitrary constant.}$$

Similarly, if we find the eigen vector corresponding to eigen value λ = 6, it will be

$$\therefore X_2 = C_2\begin{pmatrix}1\\2\end{pmatrix},\ C_2 \text{ is arbitrary constant.}$$

Note 1: The sum of the diagonal of the matrix A = sum of the eigen values of A =10. Determinant of the matrix A = product of the eigen values=24.

Note 2: If all the *n* eigen values of *A* are distinct, then there corresponds *n* distinct linearly independent eigen vectors.

Example 2: Find the eigen values and eigen vectors of

$A^T = \begin{pmatrix}8 & 2\\-4 & 2\end{pmatrix}$ **(Transpose of A) [Problem related to property- 4)]**

Solution: Characteristic equation is $\begin{vmatrix}8-\lambda & 2\\-4 & 2-\lambda\end{vmatrix} = 0,\ \Rightarrow \lambda^2 - 10\lambda + 24 = 0\ \Rightarrow \lambda = 4, 6.$

Similarly, we can find the eigen vectors corresponding to the eigen values as above. However, the eigen vectors are not the same and the corresponding eigen vectors are

$\lambda = 4,\quad X_1 = C_1\begin{pmatrix}-2\\1\end{pmatrix}$ and $\lambda = 6,\quad X_2 = C_2\begin{pmatrix}1\\-1\end{pmatrix}$. C_1 and C_2 are arbitrary constants.

Note: Eigen values of both the matrices A and A^T are the 4 and 6

Example 3: Find the eigen values and eigen vectors of

$A^{-1} = \frac{1}{24}\begin{pmatrix}2 & 4\\-2 & 8\end{pmatrix}$ **(Inverse of A) [Problem related to property- 5)]**

Solution: Characteristic equation of A^{-1} is $\left|A^{-1} - \lambda I\right| = 0$

$$\Rightarrow 24\lambda^2 - 10\lambda + 1 = 0 \qquad \Rightarrow \lambda = \frac{1}{4}, \frac{1}{6}$$

Therefore, the eigen values of A^{-1} are $\frac{1}{4}, \frac{1}{6}$ which are the reciprocals of the eigen values 4, 6 of A. Similarly, we can find the eigen vectors corresponding to the eigen values as above.

$$\lambda = \frac{1}{4}, \quad X_1 = C_1\begin{pmatrix}1\\1\end{pmatrix} \text{ and } \lambda = \frac{1}{6}, \quad X_2 = C_2\begin{pmatrix}1\\2\end{pmatrix}$$

Note: Eigen values of A^{-1} are reciprocal of the eigen values of A

Example 4: Find the eigen values and eigen vectors of

$$A^2 = \begin{pmatrix}56 & -40\\20 & -4\end{pmatrix}$$ **(Square matrix of A) [Problem related to property- 5)]**

Solution: Characteristic equation of A^2 is $\left|A^2 - \lambda I\right| = 0$

$$\Rightarrow \lambda^2 - 52\lambda + 576 = 0 \quad \Rightarrow \lambda = 16,\ 36$$

Similarly, we can find the eigen vectors corresponding to the eigen values as follows:

$$\lambda = 16, \quad X_1 = C_1\begin{pmatrix}1\\1\end{pmatrix} \text{ and } \lambda = 36, \quad X_2 = C_2\begin{pmatrix}1\\2\end{pmatrix}$$

Note: Eigen values of A^2 are the square of the eigen values of A

Example 5: Find the eigen values and eigen vectors of

$$A = \begin{pmatrix}3 & 1 & 4\\0 & 2 & 6\\0 & 0 & 5\end{pmatrix}$$

Solution: For upper triangular, lower triangular and diagonal matrices, the eigen values are given by the diagonal elements. The characteristic equation is

$$\left|A - \lambda I\right| = \begin{vmatrix}3-\lambda & 1 & 4\\0 & 2-\lambda & 6\\0 & 0 & 5-\lambda\end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(2-\lambda)(5-\lambda) = 0 \qquad \Rightarrow \lambda = 2, 3, 5$$

So eigen values of A are 3, 2, 5 which are the diagonal elements of A.

For the eigen value $\lambda = 3$, we have

$$\begin{bmatrix}0 & 1 & 4\\0 & -1 & 6\\0 & 0 & 2\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = 0 \quad i.e., \quad x_2 + 4x_3 = 0,\ -x_2 + 6x_3 = 0,\ 2x_3 = 0$$

$x_2 = 0,\ x_3 = 0,\ x_1 = \text{arbitrary}$, $X_1 = C_1[1 \quad 0 \quad 0]^t$. Similarly, for $\lambda = 2$, eigen vector $X_2 = C_2[1 \quad -1 \quad 0]^t$ and for $\lambda = 5$, eigen vector $X_3 = C_3[3 \quad 2 \quad 1]^t$.

Example 6: Find the eigen values and eigen vectors of

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$$

Solution: (Hints:)

The characteristic equation is $\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

$\Rightarrow \lambda = 1,\ 2, 3$

For $\lambda_1 = 1,\ X_1 = [1 \quad -1 \quad 0]^t$

For $\lambda_2 = 2, X_2 = [2 \quad -1 \quad -2]^t$

For $\lambda_3 = 3,\ X_3 = [1 \quad 1 \quad -2]^t$

Algebraic and Geometric multiplicity

- **Algebraic multiplicity** of an eigen value λ of a square matrix A is the order of the eigen value as a root of the characteristic polynomial (i.e., if λ is a double root then algebraic multiplicity is 2).
- **Geometric multiplicity** of λ is the number of linearly independent eigen vectors corresponding to λ.

Solved Problems

Example 7:

$$A = \begin{pmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{pmatrix}$$. **Determine the algebraic and geometric multiplicity of the eigen values.**

Solution: The characteristic equation is

$$|A - \lambda I| = \lambda^3 - 5\lambda^2 + 8\lambda - 4 = 0$$

$$\Rightarrow (\lambda-1)(\lambda-2)^2 = 0$$

So, $\lambda = 1, 2, 2$ are the eigen values with $\lambda = 2$ repeated twice (double root) of multiplicity 2.

Therefore, the algebraic multiplicity of the eigen value $\lambda = 2$ is 2 and the algebraic multiplicity of the eigen value $\lambda = 1$ is 1.

Now for $\lambda = 1$, we have

$$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{matrix} x_2 = -x_3 \\ x_1 = -x_3 \end{matrix} \quad \Rightarrow X_1 = C_1\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Now for $\lambda = 2$, we have

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \Rightarrow \begin{matrix} x_3 = 0 \\ x_1 = 2x_2 \end{matrix} \quad \Rightarrow X_2 = C_2\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Thus only one eigen vector X_2 corresponds to the repeated eigen value λ = 2. The geometric multiplicity of eigen value λ = 2 is one.

Note : For an eigen value of *A*, repeated (twice or more), there may correspond one or several linearly independent eigen vectors. Thus, the set of eigen vectors may or may not form a set of *n* linearly independent vectors.

Example 8: For $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$, determine the algebraic and geometric multiplicity.

Solutions: The eigen values of the given matrix A is 1,1,3. So the algebraic multiplicity of eigen value λ = 3 is one and for λ = 1, it is two.

Now, for λ = 3, $X_1 = C_1\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ (Verify!)

For λ = 1, we have

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \quad \begin{matrix} \Rightarrow x_1 + x_2 + x_3 = 0 \\ \Rightarrow x_1 = -x_2 - x_3 \end{matrix}$$

For a choice of $x_2 = 0$, x_3 = arbitrary, the eigen vector is $X_2 = C_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$

For a choice of $x_3 = 0$, x_2 = arbitrary, the eigen vector is $X_3 = C_3 \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$. Thus, for the repeated eigen value λ = 1, there corresponds two linearly independent eigen vectors X_2 and X_3. So the geometric multiplicity of eigen value λ = 1 is two.

Assignment-I

1. Find the eigen values and eigen vectors of the following matrices:

$$(i)\begin{pmatrix} -5 & 2 \\ 2 & -2 \end{pmatrix} \quad (ii)\begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} \quad (iii)\begin{pmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{pmatrix} \quad (iv)\begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix} \quad (v)\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{pmatrix}$$

$$(vi)\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} \quad (vii)\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$Ans: (i)\,\lambda = -1, -6; \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad (ii)\,\lambda = 2, -1; \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad (iii)\,\lambda = 1,1,1; \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix},$$

$$(iv)\,\lambda = 0,3,15; \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \quad (v)\ \lambda = 1,1,1; \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad (vi)\,\lambda = 5,1,1; \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

2. Find the sum and the product of the eigenvalues of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}. \quad \textit{(Ans: 18 and 0)}$$

3. Find the eigen values and eigen vectors of

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Are the eigen vectors linearly independent? Are they orthogonal?

(Ans. $\lambda = 0, 3, 15; \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$)

4. Two of the three eigen values of a 3×3 matrices are -1, 2. If the determinant of the matrix is 4, find the third eigen value. (Ans. -2)

5. $A = \begin{bmatrix} 1 & 0 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 5 & 6 \end{bmatrix}$ are two matrices. Find the eigen values of AB and BA. Are the values are same ?

6. If $A = \begin{bmatrix} x & 4x \\ 2 & y \end{bmatrix}$ has eigen values 5 and -1, then find the values of x and y.

   (Ans. $x = -5, y = 9$ and $x = 1,\ y = 3$)

7. Find the characteristic roots of $A^{30} - 9A^{28}$, where $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. (Ans. $0, -8$)

8. Find the eigen values of $A^2 + 2I_3$, where $A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -2 & 0 \\ 3 & 5 & 3 \end{bmatrix}$ and I is the unit matrix of order 3. (Ans. $3, 6, 11$)

9. If $\lambda_1, \lambda_2, \lambda_3$ are the eigen values of the matrix $\begin{bmatrix} -2 & -9 & 5 \\ -5 & -10 & 7 \\ -9 & -21 & 14 \end{bmatrix}$. Then find $\lambda_1 + \lambda_2 + \lambda_3$ and $\lambda_1 \lambda_2 \lambda_3$.

Diagonalization

Similar Matrices: A is said to be similar to B if there exists a non-singular matrix P such that $B = P^{-1}AP$. This transformation of A to B is known as similarity transformation. Similar matrices A and B have same eigen values. Further if X is an eigen vector of A then $Y = P^{-1}X$ is an eigen vector of the matrix B.

Diagonalization: A n-square matrix A with n linearly independent eigen vectors is similar to a diagonal matrix D whose diagonal elements are the eigen values of A. i.e $D = P^{-1}AP$

Now, $D^2 = (P^{-1}AP)(P^{-1}AP) = P^{-1}A(PP^{-1})AP = P^{-1}AIAP = P^{-1}A^2P$

Therefore, $D^2 = P^{-1}A^2P$

Similarly, $D^3 = P^{-1}A^3P$

Thus, $D^n = P^{-1}A^nP$

To obtain A_n, pre-multiply by P and post-multiply by P^{-1}, we get

$$\begin{aligned} PD^nP^{-1} &= P(P^{-1}A^nP)P^{-1} = (PP^{-1})A^n(PP^{-1}) \\ &= IA^nI = A^n \end{aligned}$$

Therefore, $A^n = PD^nP^{-1}$

Solved Problems

Examples 1: Find a matrix P which diagonalizes the matrix

$A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ **. Verify that** $P^{-1}AP = D$ **. where D is the diagonal matrix. Hence find** A^6 **.**

Solution: A is diagonalizable by P whose columns are the linearly independent eigen vectors of A. The characteristic equation of A is

$$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0 \quad \Rightarrow \lambda^2 - 7\lambda + 10 = 0 \quad \Rightarrow \lambda = 2, 5$$

For $\lambda = 2$, $2x_1 + x_2 = 0$, $x_2 = -2x_1$, $X_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$

For $\lambda = 5$, $-x_1 + x_2 = 0$, $x_2 = x_1$, $X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$P=[X_1,\ X_2]=\begin{bmatrix}1 & 1\\ -2 & 1\end{bmatrix}$$

Verification:

$$P^{-1}AP=\frac{1}{3}\begin{bmatrix}1 & -1\\ 2 & 1\end{bmatrix}\begin{bmatrix}4 & 1\\ 2 & 3\end{bmatrix}\begin{bmatrix}1 & 1\\ -2 & 1\end{bmatrix}$$

$$=\frac{1}{3}\begin{bmatrix}2 & -2\\ 10 & 5\end{bmatrix}\begin{bmatrix}1 & 1\\ -2 & 1\end{bmatrix}=\frac{1}{3}\begin{bmatrix}6 & 0\\ 0 & 15\end{bmatrix}$$

$$=\begin{bmatrix}2 & 0\\ 0 & 5\end{bmatrix}=D$$, a diagonal matrix containing eigen values 2, 5 as diagonal elements.

To find A^6, we have

$$A^6=PD^6P^{-1}=\begin{bmatrix}1 & 1\\ -2 & 1\end{bmatrix}\begin{bmatrix}2^6 & 0\\ 0 & 5^6\end{bmatrix}\frac{1}{3}\begin{bmatrix}1 & -1\\ 2 & 1\end{bmatrix}$$

$$\Rightarrow A^6=\frac{1}{3}\begin{bmatrix}64 & 15625\\ -128 & 15625\end{bmatrix}\begin{bmatrix}1 & -1\\ 2 & 1\end{bmatrix}=\frac{1}{3}\begin{bmatrix}31314 & 15561\\ 31122 & 15753\end{bmatrix}$$

$$\therefore A^6=\begin{bmatrix}10438 & 5187\\ 10374 & 5251\end{bmatrix}$$

Example 2: Diagonalize the matrix $A=\begin{pmatrix}1 & 6 & 1\\ 1 & 2 & 0\\ 0 & 0 & 3\end{pmatrix}$ and then find A^8. Also, find the modal matrix.

Solution: The characteristic equation of A is

$$|A-\lambda I|=\begin{vmatrix}1-\lambda & 6 & 1\\ 1 & 2-\lambda & 0\\ 0 & 0 & 3-\lambda\end{vmatrix}=0$$

$$\Rightarrow(\lambda+1)(\lambda-3)(\lambda-4)=0$$

$$\Rightarrow \lambda=-1,3,4$$

So eigen values of A are $\lambda=-1,\ 3,\ 4$

$For \quad \lambda = -1,$

$2x_1 + 6x_2 + x_3 = 0, \;\; x_1 + 3x_2 = 0, \;\; 4x_3 = 0$

$\Rightarrow x_3 = 0, \;\; x_1 = -3x_2 \;\therefore X_1 = C_1\begin{bmatrix}-3 & 1 & 0\end{bmatrix}^t$

$For \quad \lambda = 3,$

$-2x_1 + 6x_2 + x_3 = 0, \;\; x_1 - x_2 = 0$

$\Rightarrow x_1 = x_2, \;\; x_3 = -4x_2, \;\therefore X_2 = C_2\begin{bmatrix}1 & 1 & -4\end{bmatrix}^t$

$For \quad \lambda = 4,$

$-3x_1 + 6x_2 + x_3 = 0, \; x_1 - 2x_2 = 0, \; -x_3 = 0$

$\Rightarrow x_3 = 0, \;\; x_2 = 2x_2, \;\therefore X_3 = C_3\begin{bmatrix}2 & 1 & 0\end{bmatrix}^t$

Thus,

$$P = \begin{bmatrix}-3 & 1 & 2\\ 1 & 1 & 1\\ 0 & -4 & 0\end{bmatrix} \text{ and } P^{-1} = \frac{1}{20}\begin{bmatrix}-4 & 8 & 1\\ 0 & 0 & -5\\ 4 & 12 & 4\end{bmatrix} \text{ (Verify!)}$$

$$D = P^{-1}AP = \frac{1}{20}\begin{bmatrix}-4 & 8 & 1\\ 0 & 0 & -5\\ 4 & 12 & 4\end{bmatrix}\begin{bmatrix}1 & 6 & 1\\ 1 & 2 & 0\\ 0 & 0 & 3\end{bmatrix}\begin{bmatrix}-3 & 1 & 2\\ 1 & 1 & 1\\ 0 & -4 & 0\end{bmatrix} = \begin{bmatrix}-1 & 0 & 0\\ 0 & 3 & 0\\ 0 & 0 & 4\end{bmatrix}$$

Assignment-II

1. Diagonalize the following matrices:

   $(i)\begin{pmatrix}5 & 3\\ 3 & 5\end{pmatrix} \quad (ii)\begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix} \quad (iii)\begin{pmatrix}5 & 4\\ 1 & 2\end{pmatrix} \quad (iv)\begin{pmatrix}1 & 1 & 3\\ 1 & 5 & 1\\ 3 & 1 & 1\end{pmatrix} \quad (v)\begin{pmatrix}7 & -2 & 1\\ -2 & 10 & -2\\ 1 & -2 & 7\end{pmatrix}$

2. If A is similar to B then prove that A^2 is similar to B^2.
3. If A and B are similar then prove that they have same eigenvalues.
4. Find a matrix P which diagonalize the matrix $A = \begin{bmatrix}4 & 1\\ 2 & 3\end{bmatrix}$. Verify $P^{-1}AP = D$, where D is the diagonal matrix.
5. Show that the matrix $A = \begin{bmatrix}8 & -6 & 2\\ -6 & 7 & -4\\ 2 & -4 & 3\end{bmatrix}$ is diagonalizable. Find the transforming matrix and the diagonal matrix.